

PHY201 2.0

Optics

Text Book

FUNDAMENTALS OF OPTICS Fourth Edition

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 McGraw-Hill
Custom Publishing

New York St. Louis San Francisco Auckland Bogotá
Cairo Lisbon London Madrid Mexico Milan Montreal
New Delhi Paris San Juan Singapore Sydney Tokyo Toronto

Optics: Classifications

Geometrical optics

- Rectilinear propagation
- Finite speed
- Reflection
- Refraction
- Dispersion

Wave optics

- Interference
- Diffraction
- Electromagnetic character
- Polarization
- Double refraction

Quantum optics

- Atomic orbits
- Probability densities
- Energy levels
- Quanta
- Lasers

1.1 THE RECTILINEAR PROPAGATION OF LIGHT

The rectilinear propagation of light is the technical terminology applied to the principle that "light travels in straight lines." The fact that objects can be made to cast fairly sharp shadows may be considered a good demonstration of this principle.

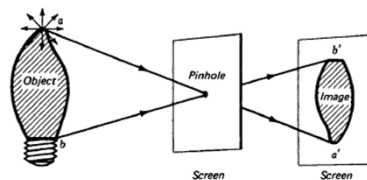


FIGURE 1A
A demonstration experiment illustrating the principle that light rays travel in straight lines. The rectilinear propagation of light.

1.2 THE SPEED OF LIGHT

The ancient astronomers believed that light traveled with an infinite speed. Any major event that occurred among the distant stars was believed to be observable instantly at all other points in the universe.

It is said that around 1600 Galileo tried to measure the speed of light but was not successful. He stationed himself on a hilltop with a lamp and his assistant on a distant hilltop with another lamp. The plan was for Galileo to uncover his lamp at an agreed signal, thereby sending a flash of light toward his assistant. Upon seeing the light the assistant was to uncover his lamp, sending a flash of light back to Galileo,

In 1849 the French physicist Fizeau* became the first man to measure the speed of light here on earth. His apparatus is believed to have looked like Fig. 1C. His account of this experiment is quite detailed, but no diagram of his apparatus is given in his notes.

Since the wheel contained 720 teeth, Fizeau found the maximum intensity to occur when its speed was 25 rev/s. The time required for each light pulse to travel over and back could be calculated by $(\frac{1}{25})(\frac{1}{2}) = 1/18,000$ s. From the measured distance over and back of 17.34 km, this gave a speed of

$$v = \frac{d}{t} = \frac{17.34 \text{ km}}{1/18,000 \text{ s}} = 312,000 \text{ km/s}$$

Electromagnetic waves of all wavelengths, from X rays at one end of the spectrum to the longest radio waves, are believed to travel with exactly the same speed

the most generally accepted value of this universal constant,

$$c = 299,792.5 \text{ km/s} = 2.997925 \times 10^8 \text{ m/s}$$

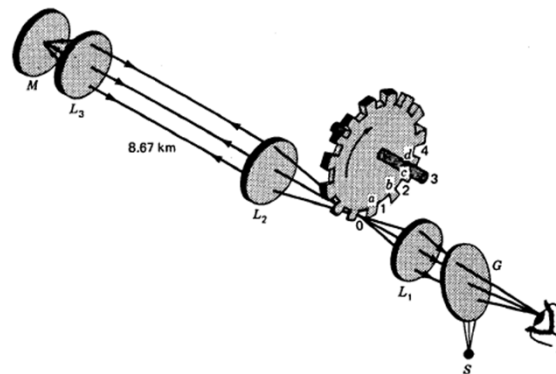


FIGURE 1C
Experimental arrangement described by the French physicist Fizeau, with which he determined the speed of light in air in 1849.

1.3 THE SPEED OF LIGHT IN STATIONARY MATTER

In 1850, the French physicist Foucault* completed and published the results of an experiment in which he had measured the speed of light in water. Foucault's experiment was of great importance for it settled a long controversy over the nature of light.

Newton and his followers in England and on the Continent believed light to be made up of small particles emitted by every light source. The Dutch physicist Huygens, on the other hand, believed light to be composed of waves, similar to water or sound waves.

Foucault's apparatus for this experiment is shown in Fig. 1D. Light coming through a slit S is reflected from a plane rotating mirror R to the equidistant concave mirrors M_1 and M_2 . When R is in the position 1, the light travels to M_1 , back along the same path to R , through the lens L , and by reflection to the eye at E . When R is in position 2, the light travels the lower path through an auxiliary lens L' and tube T to M_2 , back to R , through L to G , and then to the eye E . If now the tube T is filled with water and the mirror is set into rotation, there will be a displacement of the images from E to E_1 and E_2 . Foucault observed that the light ray through the tube was more displaced than the other. This means that it takes the light longer to travel the lower path through water than it does the upper path through the air.

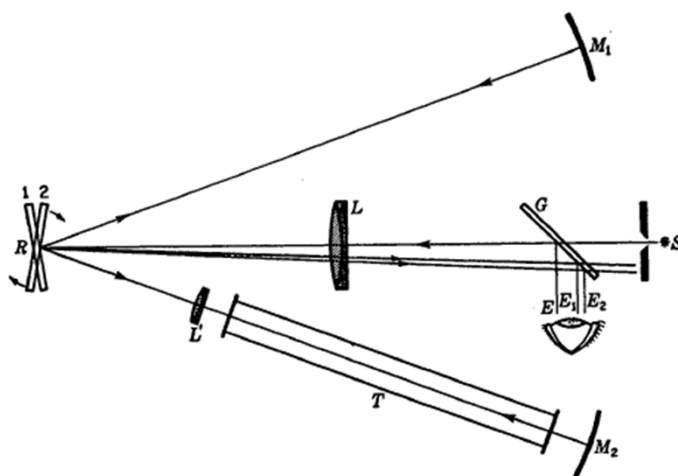


FIGURE 1D
Foucault's apparatus for determining the speed of light in water.

Over 40 years later the American physicist Michelson (first American Nobel laureate 1907) measured the speed of light in air and water. For water he found the value of 225,000 km/s, which is just three-fourths the speed in a vacuum. In ordinary optical glass, the speed was still lower, about two-thirds the speed in a vacuum.

1.4 THE REFRACTIVE INDEX

The index of refraction, or refractive index, of any optical medium is defined as the ratio between the speed of light in a vacuum and the speed of light in the medium:

$$\text{Refractive index} = \frac{\text{speed in vacuum}}{\text{speed in medium}} \quad (1c)$$

$$n = \frac{c}{v}$$

The *optical density* of any transparent medium is a measure of its refractive index. A medium with a relatively high refractive index is said to have a high optical density, while one with a low index is said to have a low optical density.

1.5 OPTICAL PATH

To derive one of the most fundamental principles in geometric optics, it is appropriate to define a quantity called the *optical path*. The path d of a ray of light in any medium is given by the product *velocity times time*:

$$d = vt$$

Since by definition $n = c/v$, which gives $v = c/n$, we can write

$$d = \frac{c}{n} t \quad \text{or} \quad nd = ct$$

The product nd is called the *optical path* Δ :

$$\Delta = nd$$

The optical path represents the distance light travels in a vacuum in the same time it travels a distance d in the medium. If a light ray travels through a series of optical media of thickness $d, d', d'', \dots$ and refractive indices $n, n', n'', \dots$, the total optical path is just the sum of the separate values:

$$\bullet \quad \Delta = nd + n'd' + n''d'' + \dots$$

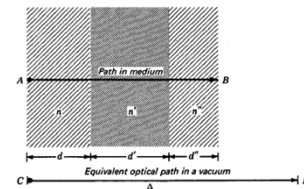


FIGURE 1E
The optical path through a series of optical media.

1.6 LAWS OF REFLECTION AND REFRACTION

Whenever a ray of light is incident on the boundary separating two different media, part of the ray is reflected back into the first medium and the remainder is refracted (bent in its path) as it enters the second medium (see Fig. 1F). The directions taken by these rays can best be described by two well-established laws of nature.

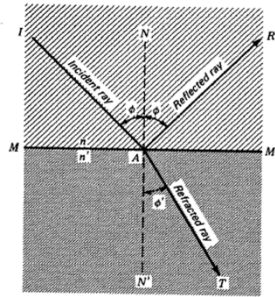


FIGURE 1F
Reflection and refraction at the boundary separating two media with refractive indices n and n' , respectively.

$$\text{angle of incidence} = \text{angle of reflection} \quad (1j)$$

$$\frac{\sin \phi}{\sin \phi'} = \text{const} \quad (1k)$$

Furthermore, the refracted ray also lies in the plane of incidence and on the opposite side of the normal. This relationship, experimentally established by Snell,* is known as *Snell's law*. In addition the constant is found to have exactly the ratio of the refractive indices of the two media n and n' . Hence we can write

$$\frac{\sin \phi}{\sin \phi'} = \frac{n'}{n} \quad (1l)$$

$$\frac{\sin \phi}{\sin \phi'} = \frac{v}{v'} \quad (1o)$$

$$\frac{\sin \phi}{\sin \phi'} = n' \quad (1p)$$

etc. When the angles of incidence and refraction are very small, a good approximation is obtained by setting the sines of angles equal to the angles themselves, obtaining

$$\frac{\phi}{\phi'} = \frac{n'}{n} \quad (1q)$$

1.7 GRAPHICAL CONSTRUCTION FOR REFRACTION

A simple method for tracing a ray of light across a boundary separating two optically transparent media is shown in Fig. 1G. Because the principles involved in this construction are readily extended to complicated optical systems, the method is useful in the preliminary design of many different kinds of optical instruments.

the *angle of deviation*, is given by

$$\beta = \phi - \phi' \quad (1r)$$

To prove that this construction follows Snell's law exactly, we apply the law of sines to the triangle *ORP*:

$$\frac{OR}{\sin \phi'} = \frac{OP}{\sin (\pi - \phi)}$$

Since $\sin (\pi - \phi) = \sin \phi$, $OR = n$, and $OP = n'$, substitution gives directly

$$\frac{n}{\sin \phi'} = \frac{n'}{\sin \phi} \quad (1s)$$

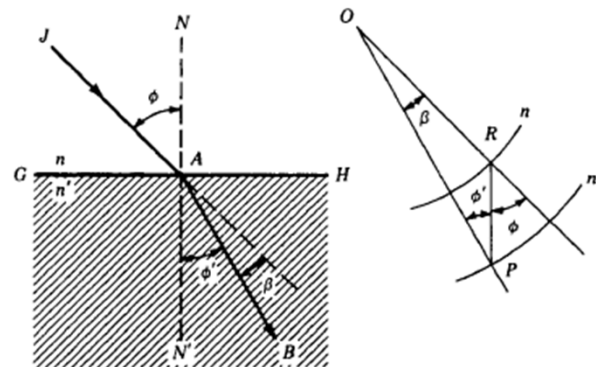


FIGURE 1G
Graphical construction for refraction at a smooth surface separating two media of index n and n' .

1.8 THE PRINCIPLE OF REVERSIBILITY

The symmetry of Eqs. (1j) and (1m) with respect to the symbols used shows at once that if a reflected or refracted ray is reversed in direction, it will retrace its original path. For any given pair of media with indices n and n' any one value of ϕ is correlated with a corresponding value of n' . This will be equally true when the ray is reversed and ϕ' becomes the angle of incidence in the medium of n' ; the angle of refraction will then be ϕ . Since reversibility holds at each reflecting and refracting surface, it holds also for even the most complicated light paths. This useful principle has more than a purely geometrical foundation, and later it will be shown to follow from the application of wave motion to a principle in mechanics.

1.9 FERMAT'S PRINCIPLE

The path taken by a light ray in going from one point to another through any set of media is such as to render its optical path equal, in the first approximation, to other paths closely adjacent to the actual one.

The other paths must be possible ones in the sense that they may undergo deviations only where there are reflecting or refracting surfaces. Fermat's principle will hold for a ray whose optical path is a minimum with respect to adjacent hypothetical paths. Fermat himself stated that the time required by the light to traverse the path is a minimum and the optical path is a measure of this time. But there are plenty of cases in which the optical path is a maximum or neither a maximum nor a minimum but merely stationary (at a point of inflection) at the position of the true ray.

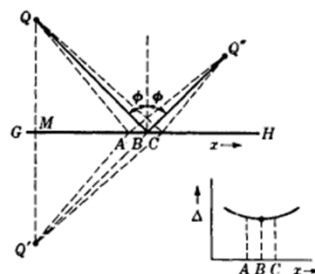


FIGURE 1I
Fermat's principle applied to reflection
at a plane surface.

Consider now adjacent paths to points like A and C on the mirror surface close to B . Since a straight line is the shortest path between two points, both the paths $Q'AQ''$ and $Q'CQ''$ are greater than $Q'BQ''$. By the above construction and equivalent triangles, $QA = Q'A$, and $QC = Q'C$, so that $QAQ'' > QBQ''$ and $QCQ'' > QBQ''$. Therefore the real path QBQ'' is a minimum. A graph of hypothetical paths close to the real path QBQ'' , as shown in the lower right of the diagram, indicates the meaning of a minimum, and the flatness of the curve between A and C illustrates that to a first approximation adjacent paths are equal to the real optical path.

Consider finally the optical properties of an ellipsoidal reflector, as shown in Fig. 1J. All rays emanating from a point source Q at one focus are reflected according to the law of reflection and come together at the other focus Q' . Furthermore all paths are equal in length.

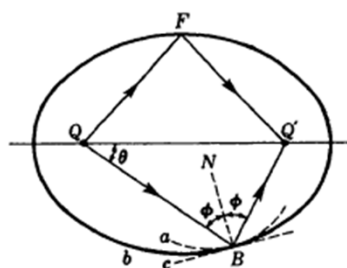
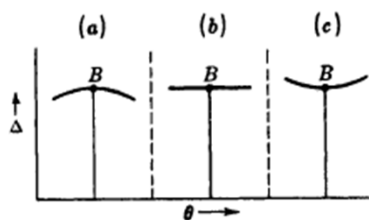


FIGURE 1J
Fermat's principle applied to an elliptical reflector.

FIGURE 1K
Graphs of optical paths involving reflection illustrating conditions for (a) maximum, (b) stationary, and (c) minimum light paths. Fermat's principle.



Adjacent paths from Q to points along these mirrors, however, will give a minimum condition for the real path to and from reflector c and a maximum condition for the real path to and from reflector a (see Fig. 1K).

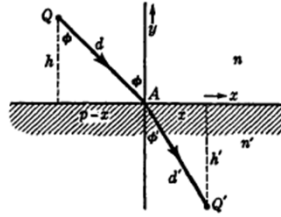


FIGURE 1L
Geometry of a refracted ray used in
illustrating Fermat's principle.

It is readily shown mathematically that both the laws of reflection and refraction follow from Fermat's principle. Figure 1L, which represents the refraction of a ray at a plane surface, can be used to prove the law of refraction [Eq. (1m)]. The length of the optical path between the point Q in the upper medium of index n and another point Q' in the lower medium of index n' passing through any point A on the surface is

$$\Delta = nd + n'd' \quad (1u)$$

where d and d' represent the distances QA and AQ' , respectively.

Now if we let h and h' represent perpendicular distances to the surface and p the total length of the x axis intercepted by these perpendiculars, we can invoke the pythagorean theorem concerning right triangles and write

$$d^2 = h^2 + (p - x)^2 \quad d'^2 = h'^2 + x^2$$

$$\Delta = n[h^2 + (p - x)^2]^{1/2} + n'[h'^2 + x^2]^{1/2} \quad (1v)$$

According to Fermat's principle, Δ must be a minimum or a maximum

By differentiating Eq. (1v) with respect to x and setting the result equal to zero, we obtain

$$\frac{d\Delta}{dx} = \frac{\frac{1}{2}n}{[h^2 + (p - x)^2]^{1/2}} (-2p + 2x) + \frac{\frac{1}{2}n'}{(h'^2 + x^2)^{1/2}} 2x = 0$$

which gives

$$n \frac{p - x}{[h^2 + (p - x)^2]^{1/2}} = n' \frac{x}{(h'^2 + x^2)^{1/2}}$$

or simply

$$n \frac{p - x}{d} = n' \frac{x}{d'}$$

$$n \sin \phi = n' \sin \phi' \quad (1w)$$

1.10 COLOR DISPERSION

the incident ray of white light gives rise to refracted rays of different colors (really a continuous *spectrum*) each of which has a different value of ϕ' . By Eq. (1m) the value of n' must therefore vary with color. It is customary in the exact specification of indices of refraction to use the particular colors corresponding to certain dark lines in the spectrum of the sun. These *Fraunhofer** lines, which are designated by the letters A, B, C, . . . , starting at the extreme red end, are given in Table 1A. The ones most commonly used are those in Fig. 1M.

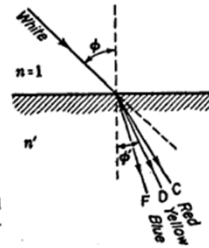


FIGURE 1M
Upon refraction, white light is spread out into a spectrum. This is called dispersion.

typical case of crown glass, the refractive indices as given in Table 1A are

$$n_F = 1.52933 \quad n_D = 1.52300 \quad n_C = 1.52042$$

Now it is readily shown from Eq. (1q) that for a given small angle ϕ the dispersion of the F and C rays ($\phi_F - \phi_C$) is proportional to

$$n_F - n_C = 0.00891$$

while the deviation of the D ray ($\phi - \phi_D$) depends on $n_D - 1$ which is equal to 0.52300. Thus it is nearly 60 times as great. The ratio of these two quantities varies greatly for different kinds of glass and is an important characteristic of any optical substance. It is called the *dispersive power* and is defined by the equation

$$V = \frac{n_F - n_C}{n_D - 1} \quad (1x)$$

The reciprocal of the dispersive power is called the *dispersive index* v :

$$v = \frac{n_D - 1}{n_F - n_C} \quad (1y)$$

For most optical glasses v lies between 20 and 60

Table 1B DISPERSION INDEX FOR FOUR OPTICAL GLASSES*

Glass	Spectacle crown	Light flint	Dense flint	Extra dense flint
v	58.7	41.2	47.6	29.08